Chapter 4 — An End-to-End Empirical Study of Solana ↔ Ethereum Cross-Chain Arbitrage

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1 LANDSCAPE OF SOLANA ↔ ETHEREUM BRIDGES

Five general-purpose bridges currently support sustained asset transfers between the Solana and Ethereum ecosystems: Wormhole Portal, LayerZero V2, Mayan Finance, Relay, and Chainlink CCIP. They differ substantively in trust model, fee schedule, and verification mechanism, resulting in distinct latency, fee, and per-transfer-cap profiles. Table 1 summarises their main parameters alongside the volume of bridge events we were able to collect for each.

Table 1: Comparison of SOL ↔ ETH bridges

Protocol	Trust / Verification	Fee model	Records
Wormhole Portal	Guardian-signed VAA	Gas-based	42,181
LayerZero V2	DVN verification	DVN fees	271,813
Mayan Finance	Dutch auction	~3 bps	10,653,600
Relay	Intent-based fill	Variable bps	107,220
Chainlink CCIP	DON + risk manager	LINK fees	2,115

Among these, Wormhole and CCIP adopt an attester-based "messaging layer + token wrapping" architecture; LayerZero is layered across DVN attestation and Oracle/Relayer verification; Mayan and Relay model cross-chain settlement as an "intent/auction" in which solvers compete to fill orders. These architectural differences determine: (i) whether cross-chain events carry on-chain, 1:1-verifiable cryptographic fingerprints; (ii) whether each transfer can be paired into an (src, dst) tuple through a single public field; and (iii) whether there is intermediary-priced latency–fee heterogeneity. These three properties are exactly the preconditions for reconstructing per-candidate arbitrage PnL in the remainder of the chapter.

2 WHY WORMHOLE: DATA AND CANDIDATE CONSTRUCTION

2.1 Why Wormhole Portal

Of the five bridges above, we use Wormhole Portal as our sole long-horizon sample source. Our reasoning is fourfold: (i) *On-chain verifiability.* Every Portal message is cryptographically signed by the Guardian cluster and committed on-chain as a Verifiable Action Approval (VAA), so that a Solana signature and an Ethereum transaction hash can be paired 1:1 from cryptographic evidence alone, without any off-chain ledger; (ii) *Standardised pairing schema.* The Token Bridge contracts on both chains use fixed emitter/recipient field layouts, so the "actor" primary key ($arb_{\text{sol}}, arb_{\text{eth}}$) can be decoded directly from on-chain data; (iii) *Decoupled transport vs. execution.* Portal handles only token wrap/unwrap; any subsequent DEX execution on the destination chain is an independent on-chain event, allowing entry-swap, bridge-leg, and exit-swap to be attributed separately; (iv) *Widest token coverage.* For the long-tail memecoins that dominate observed cross-chain arbitrage, the token set supported by Wormhole (via its WTT and NTT standards) is a strict superset of the other four bridges, so "Wormhole-only" sampling introduces no material selection bias for the phenomenon we study.

2.2 Data Collection Pipeline

Figure 1 shows our five-stage collection and matching pipeline. The official Wormhole `/operations` API indexes by VAA but *only retains approximately* 7 *days of history*, which is insufficient for our year-long analysis. We therefore *invert the pipeline* and start from the on-chain Portal contracts on both legs: (1) on Ethereum and Solana, scan the Portal Token Bridge contract's full log/instruction stream and parse the `src_chain_id` and `dst_chain_id` of every bridge event; (2) for each event, use the Solana signature or Ethereum transaction hash as a query key against Wormholescan's sig/hash search endpoint to retrieve the corresponding VAA (this path is not bounded by the 7-day window of the `/operations` API); (3) merge the two-chain signatures/hashes into a unified cross-chain record keyed by the VAA identifier (`emitterChain + sequence`); (4) query Birdeye's *1-second* price endpoint for the token-USD rate at the entry and exit timestamps; (5) run a greedy attribution algorithm to assign same-wallet swaps immediately before/after the bridge event to the candidate, yielding tuples of the form (direction, sol_sig, eth_hash, entry_swaps, exit_swaps, $C^{\text{in}}_{\text{exec}}$, Π_{net}).

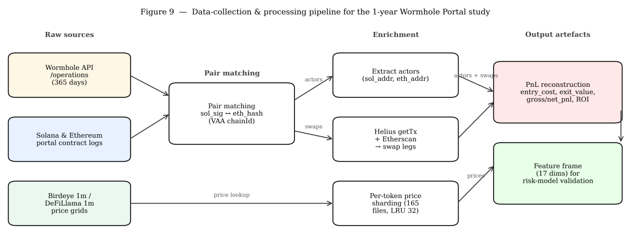

Figure 1: Data collection and arbitrage-candidate construction pipeline.

3 WHAT DRIVES CROSS-CHAIN PROFIT: DESCRIPTIVE EVIDENCE

This section answers the question: descriptively, which observables are correlated with *net PnL* Π_{net} and which are not? Figure 3 reports six sub-results at the economic level.

Distribution shape (panels a/b). ROI is heavy-tailed and left-skewed, with median +1.41% and 75.7% of samples positive; the SOL → ETH CDF has a slightly thinner left tail and ETH → SOL a slightly thicker one, but the two directional medians differ by less than 1 percentage point, so direction itself has limited explanatory power. Table 2 reports the full quantile structure for reliable candidates. The heavy-tailed profile is unambiguous: p_1 sits at −49% while p_{99} reaches +151%; the ratio std/|median| exceeds 47; the mean +7.85% is 5.5× the median +1.43%, dragged by a small number of extreme positive samples.

Table 2: ROI quantiles on reliable candidates (n = 28,788, units %)

Stat.	Value	Quantile	Value	Quantile	Value
count	28,788	p_1	−49.09	p_{75}	+3.82
mean	+7.85	p_5	−3.98	p_{90}	+10.70
std	67.24	p_{10}	−1.45	p_{95}	+27.34
min	−125.28	p_{25}	+0.26	p_{99}	+150.98
max	+5,204.09	p_{50}	+1.43	—	—

The minimum ROI of −125.28% (loss exceeding 100%) is not a calculation error: when $f_{\text{total}} > C^{\text{in}}_{\text{exec}}$, the ratio ROI = $\Pi_{\text{net}}/C^{\text{in}}_{\text{exec}}$ can mathematically cross −100%. The most extreme LOVE sample has entry \$1.00 and fees \$1.13—a genuine economic loss, not a measurement artefact.

Capital is essentially unrelated to PnL (panels c/f). On a logarithmic x-axis, entry capital spans seven decades. Fitting an OLS regression

$$\Pi_{\text{net}} = \alpha + \beta \cdot \log_{10}(C^{\text{in}}_{\text{exec}}) + \varepsilon \tag{1}$$

on the n = 28,586 reliable candidates yields $\hat{\beta}$ = −999 \$/decade with R^2 = 0.0022. In words, *entry capital explains only* 0.22% *of the variance in net PnL*. The binned-median curve in panel (f) is approximately flat and slightly positive over the 10^2–10^4 USD band and sinks at both ends—an inverted-U shape. This counter-intuitive result refutes the folk assumption that cross-chain arbitrage is a linearly capital-scalable business.

Fees are a first-order constraint (panels d/e). The fee-rate distribution is extremely wide; the median $fee/|gross|$ = 0.054, but 2.7% of candidates suffer $fee > |gross|$—severe fee erosion. In panel (e) every point lies strictly below the $y = x$ line; the vertical displacement visualises the fee cost directly, and the dense lower-left cluster corresponds to the many small-volume losses whose gross margin failed to cover gas.

Temporal structure (Fig. 2). Figure 2 characterises the time-side properties of cross-chain candidates from four angles: (a) bridge-latency CDF by direction—SOL→ETH (blue) and ETH→SOL (red) are nearly coincident, both with median ~ 800 s (~ 13 min, matching two Ethereum epochs of finality); the inter-direction gap is < 5%; (b) atomicity matrix—both-atomic (89.3%) and both-non-atomic (6.7%) dominate, while asymmetric one-side-atomic cases are small (3.9% + 0.1%); fully-atomic attempts amount to 29 candidates (0.1%); (c) swap-to-bridge time gap—the entry-side gap has a median < 12 s, the exit-side is longer (~ 24 s), with a right tail extending into the hour range (a small number of candidates hold the bridged position before exit); (d) ROI boxplot by latency tier—medians drop monotonically from the < 60 s tier through 60–300 s, 300 s–13 min, 13 min–24 h; the > 24 h tier has the smallest n and the widest dispersion. Panel (d) is the direct evidence for the "bridge latency negative-significant" row of Table 3.

Summary of "what matters / what does not." Table 3 collects the descriptive evidence above. These are all *univariate* claims; some of the variables are re-evaluated under the multivariate model in Section 4.

4 INTEGRATED MODELLING AND EXPERIMENTS

The conclusions in §3 are all univariate. If the features interact, univariate relations can mislead. This section reassesses both the direction and magnitude of Π_{net} via a multivariate non-linear model under two parallel targets: a *regression* predicting the dollar amount of Π_{net}, and a *classification* predicting $\mathbf{1}\{\Pi_{\text{net}} > 0\}$.

4.1 Feature set

We assemble 17 features covering six hypothesised driver families, all observable at the candidate level: *direction / time*

Figure 2 — Timing & Atomicity of Wormhole Portal Arbitrage

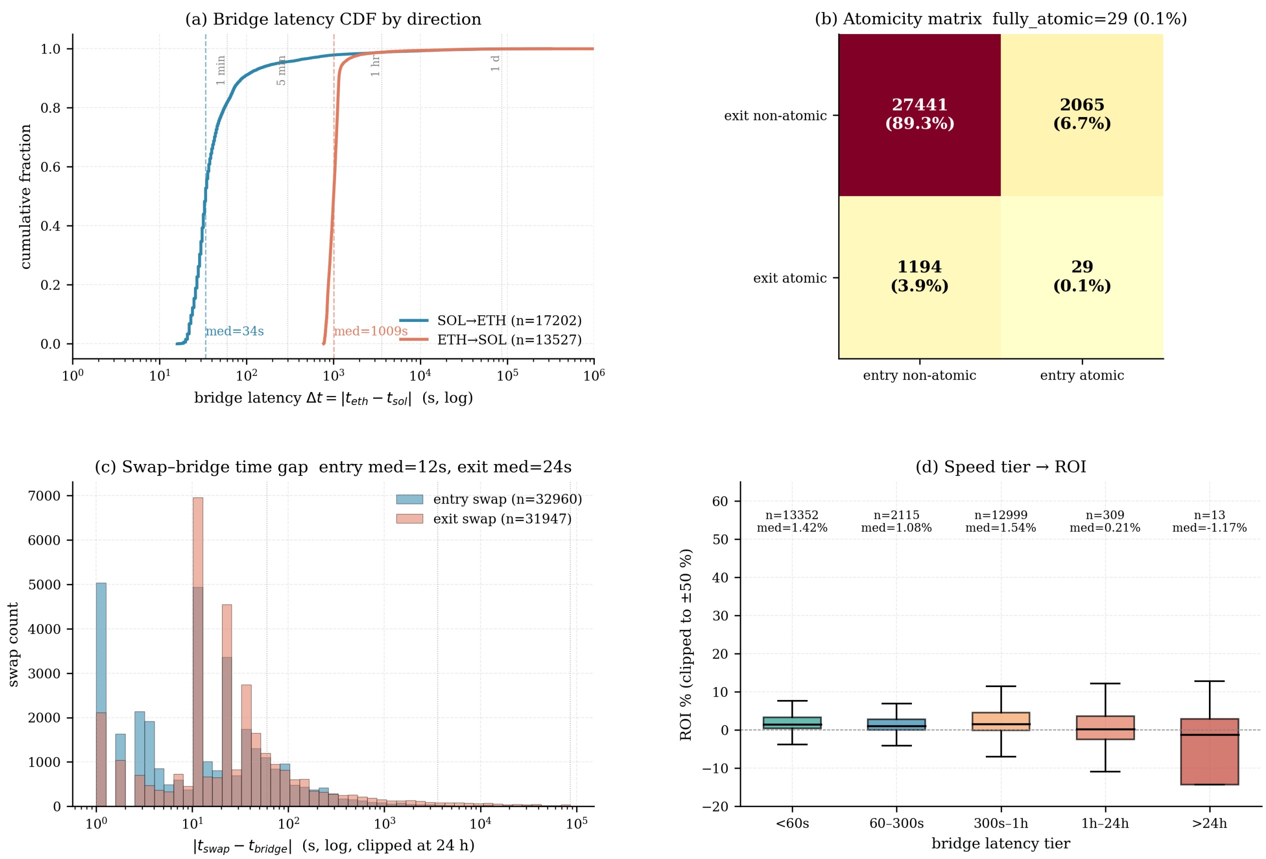

Figure 2: Temporal structure: latency CDF by direction (a), atomicity matrix (b), swap–bridge gap (c), ROI by latency tier (d).

Table 3: Descriptive relations between Π_{net} and each variable

Variable	Relation	Evidence
Entry spread	Positive, significant	Prior work + tiering
Bridge latency	Negative, significant	Fig. 2(d) + model
Liquidity depth	Non-linear	Liquidity-tier strata
Fees	Negative (identity)	Fig. 3(d/e)
Entry capital	$R^2 = 0.002$, near-null	Fig. 3(f)
Direction	Gap < 1 pp	Fig. 2(a)
Hour of day	Approximately uniform	(omitted)
Entry/exit atomicity	No inter-tier ROI gap	Fig. 2(b)
Swap split count	No significant effect	Fig. 2(c)

(dir_sol_to_eth, hour_utc); *size / latency* (log_size, log_latency); *atomicity* (atomic_entry_same_block, atomic_exit_same_block); *swap–bridge timing* (entry_delta_bridge_sec, exit_delta_bridge_sec, n_entry_swaps, n_exit_swaps); *measurement completeness* (entry_coverage, exit_coverage); *liquidity tiers and market* (liq_dead, liq_thin, liq_healthy, entry_gap_pct, actor_trade_count).

4.2 Model specification

Both models share the same additive tree ensemble:

$$F_M(x) \;=\; \sum_{m=1}^{M} \eta \cdot h_m(x), \tag{2}$$

where h_m is a depth-3 regression tree, $M = 200$, and $\eta = 0.05$. Each tree fits the negative functional gradient of the current loss:

$$h_m \;=\; \arg\min_{h} \sum_{i=1}^{n} L\big(y_i,\; F_{m-1}(x_i) + h(x_i)\big), \tag{3}$$

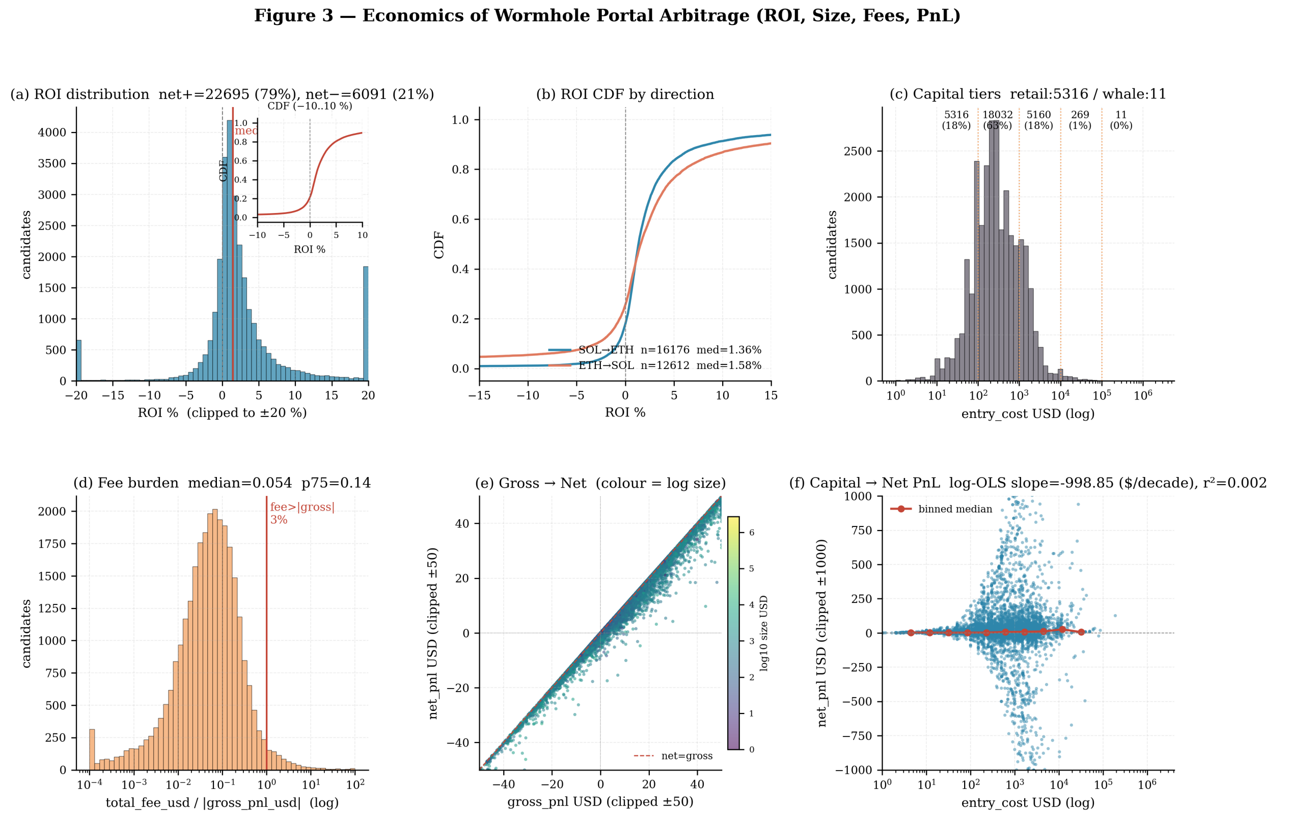

Figure 3: Economics of cross-chain arbitrage: (a) ROI histogram with inset CDF; (b) ROI CDF by direction; (c) log-scale distribution of entry capital and tier shading; (d) distribution of $fee/|gross\ PnL|$; (e) Gross vs. Net PnL scatter (colour-encoded by capital); (f) OLS regression of $\Pi_{\mathbf{net}}$ on $\log_{10}(C^{\mathbf{in}}_{\mathbf{exec}})$.

with subsample = 0.8 applied per tree to reduce variance.

Regression target ("how much"). The target is a symmetrically USD-clipped Π_{net}:

$$y_i^{\text{reg}} = \operatorname{clip}\big(\Pi_{\text{net},i},\ -\$100,\ +\$500\big), \tag{4}$$

with squared-error loss $L_{\text{reg}}(y, \hat{y}) = \frac{1}{2}(y - \hat{y})^2$. The clip bounds come from the empirical (p_1, p_{99}) of ROI: samples beyond this range load the R^2 asymmetrically and over-fit the tails.

Classification target ("whether profitable"). The target is a profit indicator:

$$y_i^{\text{cls}} = \mathbf{1}\{\Pi_{\text{net},i} > 0\}, \tag{5}$$

with binary cross-entropy $L_{\text{cls}}(y, \hat{p}) = -y \log \hat{p} - (1 - y)\log(1 - \hat{p})$, producing a probability $\hat{p}_i = \sigma\big(F_M(x_i)\big)$. A trading decision is obtained via a threshold τ: $\hat{y}_i^{\text{cls}} = \mathbf{1}\{\hat{p}_i > \tau\}$.

4.3 Evaluation protocol

The training set is restricted to candidates with `pnl_complete` and no missing features, giving n = 15,259. We use K = 5-fold shuffled KFold (`random_state= 42`): each fold trains on 4/5 of the data and tests on the held-out 1/5. Cross-validated scores are averaged across folds:

$$R^2_{\text{CV}} = \frac{1}{K}\sum_{k=1}^{K} R^2_k, \quad \text{AUC}_{\text{CV}} = \frac{1}{K}\sum_{k=1}^{K} \text{AUC}_k. \tag{6}$$

Regression reports R^2, MAE, Median AE; classification reports AUC, log-loss, and at two operating thresholds $\tau \in \{0.5, 0.8\}$,

$$\text{Precision} = \frac{TP}{TP+FP}, \quad \text{Recall} = \frac{TP}{TP+FN}. \tag{7}$$

The naive classification baseline is "always predict profit," whose accuracy equals the positive rate $\bar{y}^{\text{cls}}$ = 74.71%.

4.4 Cross-validated results

Table 4 summarises 5-fold CV performance. The regressor reaches $R^2_{\text{CV}} = 0.793 \pm 0.032$—more than two orders of magnitude above Eq. (1)'s univariate R^2 = 0.002, proving that *Π_{net} is predictable given the 17-feature set, but only under a multivariate non-linear model.* Median AE = \$5.04 implies at least half of the samples are predicted within ±$5; the gap between MAE = $16.96 and Median AE (a factor of 3.4) indicates a fat-tailed error distribution—a minority of extreme samples remain under-fit, confirming the need for the clipping in Eq. (4).

The classifier's AUC is 0.862 ± 0.006: substantially above the 0.5 random baseline but well below the 1.0 perfectly-separable limit—reflecting intrinsic randomness in cross-chain arbitrage (counterparty flow, Guardian-signing jitter) that the present feature set cannot exhaust. At $\tau = 0.5$, accuracy is 82.5%, +7.8 pp over the naive $\bar{y}^{\text{cls}} = 74.71\%$ baseline. Raising τ to 0.8 trades recall for precision: precision rises to 93.4% while recall falls to 66.8%—only trading when the model is > 0.8 confident yields a $\sim 94\%$ realised-win rate at the cost of missing 33% of the opportunities. This precision–recall schedule directly motivates using the classifier as a high-confidence strategy-side filter. Log-loss = 0.384 is also below the binomial baseline of 0.566.

Table 4: Model performance (5-fold CV, n = 15,259)

Metric	Value
Regressor R^2_{CV}	0.7934 ± 0.0318
Regressor MAE	$16.96
Regressor Median AE	$5.04
Classifier AUC_{CV}	0.8620 ± 0.0057
Classifier accuracy at $\tau = 0.5$	82.5% (baseline 74.7%)
Classifier precision at $\tau = 0.8$	93.4% (recall 66.8%)
Classifier log-loss	0.3844

4.5 Probability calibration

AUC only certifies the *ranking* of probabilities; for the $\hat{p}_i$ to be usable as a literal empirical probability (as in the $\tau = 0.8$ strategy above), the *numerical values* must also be trustworthy. We partition the pooled out-of-fold $\hat{p}$ into ten equal-frequency buckets (n_d = 1,526 each) and compare the bucket mean $\bar{\hat{p}}_d$ with the observed positive rate $\bar{y}_d = \frac{1}{n_d} \sum_{i \in d} y_i^{\text{cls}}$; see Table 5. All absolute gaps $|\bar{\hat{p}}_d - \bar{y}_d| < 5$ pp; the largest is −4.7 pp in D2, while the remaining nine bins deviate by ≤ 4 pp. For the high-confidence half of the distribution (D5–D10) the actual positive rate exceeds the predicted probability by 2–3 pp—the model is *slightly pessimistic* where it matters most for a conservative trading rule.

Table 5: Decile calibration of the classifier's 5-fold CV predicted probabilities (per-bin n = 1,526)

Bin	Pred. mean	Actual rate	Gap
D1	0.1795	0.1363	−0.0432
D2	0.5200	0.4731	−0.0469
D3	0.6642	0.6298	−0.0344
D4	0.7389	0.7051	−0.0338
D5	0.7923	0.8211	+0.0288
D6	0.8413	0.8590	+0.0177
D7	0.8892	0.9227	+0.0335
D8	0.9245	0.9574	+0.0329
D9	0.9516	0.9738	+0.0222
D10	0.9708	0.9928	+0.0220

5 LIMITATIONS

Sensitivity to the dust-filter threshold. We set the dust-filter threshold at $C^{\text{in}}_{\text{exec}} < \$1$, motivated by the $1–$5 physical floor of Ethereum L1 swap gas. Raising the threshold to $5 increases the set of excluded rows from 11 to ≈ 120 (∼ 0.4%), lowers the ROI mean on the reliable subset from +7.85% to about +4.2%, while leaving the median essentially unchanged (+1.43% → +1.45%). The GBM results of Section 4 (R^2, AUC, Median AE) do not change materially under either threshold. We keep the more conservative $1 threshold to maximise sample size and avoid discarding legitimate small-size trades where "fees exceed entry" is a real economic loss.

Single-bridge sample bias. We collect candidates only from Wormhole Portal, leaving LayerZero V2, Mayan, Relay, and CCIP out of scope. Section 1 argued that Wormhole has the broadest memecoin coverage and the largest candidate volume among the five, and that its "messaging layer + token wrapping" structure permits exact PnL reconstruction. That said, Portal's end-to-end latency is a median of ∼ 13 s (Solana finality) plus ∼ 13 min (two Ethereum epochs)—numbers specific to Wormhole. On LayerZero or intent-based Mayan/Relay, the latency mechanism, tail shape, and the arbitrage window differ substantially, so the chapter's model coefficients and R^2_{CV} should not be extrapolated to other bridges; a rigorous cross-bridge comparison is future work.

Observation window and regime shifts. The dataset spans approximately 365 days, over which Solana/Ethereum fee regimes, Wormhole's Guardian signing cadence, and the behaviour of top bots may all have undergone regime shifts. The model in this chapter is a *mixed-period* average and is not segmented; splitting the window into distinct regimes may cause both model coefficients and CV scores to drift. Regime identification and segment-wise modelling are left to future work.

6 CONCLUSION

On one year of Wormhole Portal data we construct 30,729 cross-chain arbitrage candidates and, on the 28,574 reliable subset, provide descriptive and multivariate evidence in parallel. The descriptive layer (§3) shows that both the *direction* and *magnitude* of cross-chain profit are *not* determined by the micro-structural variables commonly assumed to matter in the literature—entry capital, trade direction, hour of day, or swap atomicity—while entry spread and bridge latency are the two exogenous variables significantly correlated with Π_{net} (Table 3). The modelling layer (§4) achieves 5-fold cross-validated regression $R^2_{\text{CV}} = 0.79$, classification $\text{AUC}_{\text{CV}} = 0.86$, and Median AE of \$5, showing that Π_{net} is *predictable* under the 17-feature set; all ten decile-bucket calibration gaps are < 5 pp, letting the predicted probability feed directly into strategy-side threshold decisions—the $\tau = 0.8$ operating point yields a 93.4%-precision high-confidence filter. Two follow-ups suggest themselves: (a) the upstream determinants of latency heterogeneity (Guardian signing cadence, asymmetric chain finality); and (b) structural modelling of liquidity impact within the arbitrage window, which is essential to characterise the true edge in cross-chain arbitrage.